INSTITUTE OF TECHNOLOGY OF CAMBODIA

Student Information Management

Subject: Data Structure and Programing 2

Lecturer: Ms. CHOM Sreylam

Group: 06

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Linked List Queue

Group Members

1. Hin Panha
2. Hor Soue Hak
3. Kun Virak

Data Structure

Student Node

- id
- name
- age
- gender
- department
- major
- year
- province
- phone
- birth
- email
- → **next**

Queue (Linked List)

Student 1 → **Student 2** → **Student 3**

FRONT **REAR**

- Nodes stored in memory (heap)
- front & rear track the queue
- Each node links to the next
- CSV saves data permanently

Program Start

Open CSV File

Reads Students_Information.csv

Load into Queue

Each row → a new Student node

Show Menu

User picks an action (1–6)

Run Action

Execute the chosen function

Loop Again

Repeat until user exits (6)

Loop continues until user selects Exit (6)

Menu Options

1 Add Student

4 Search

2 Remove Student

5 Update Info

3 Display All

6 Exit

Add Student

1. Enter Student ID
2. Check if ID already exists
3. **Duplicate ID? → Stop & warn**
4. Enter all student info
5. Add node to end of queue
6. Save to CSV file

Fields Collected

ID • Name • Age

Gender • Department

Major • Year

Province • Phone

Birth • Email

Always saved to CSV after adding

Remove Search

Remove Student

- Enter Student ID to delete
- Search through the queue
- Adjust node pointers
- Free memory (delete node)
- Update REAR if needed
- Save to CSV

Search Student

- Enter Student ID to find
- Walk through queue nodes
- Compare each node's ID
- Found → show all details
- Not found → print error

Display & Sort

Bubble Sort (A → Z)

1. Compare name with next node
2. If out of order → swap ALL fields
3. Repeat until no more swaps
4. List is now sorted A → Z
5. Then display the table

Table Output

ID | Name | Age | Gender | Dept

Major | Year | Province | Phone | Birth | Email

Setw( ) aligns each column evenly

- Sorted alphabetically first
- No file save needed for display

Update Student

- Enter ID to find student
- Search queue with findStudent( )
- Not found? → show error & stop
- Show update menu (options 1–11)
- Pick which field to change
- Change saved to CSV file

11 Updatable Fields

ID	Year
Name	Province
Age	Phone
Gender	Birth
Department	Email
Major	

File Storage (CSV)

SaveFile()

- → Open CSV file for writing
- → Write header row first
- → Loop all nodes in queue
- → Write each field with commas
- → Close the file

DataIntoQueue()

- → Open CSV file for reading
- → Skip header row (getline)
- → Read comma-separated fields
- → Create new Student node
- → Add node to the queue

CSV Example

```
ID,Name,Age,Gender,Department,Major,Year,Province,Phone,Birth,Email
e20241363,Hin Panha,19,Male,AMS,Data Science,2,Battambong,088 9157 750,06/12/2006,panhaaahinn@example.com
```

Conclusion

Linked List

Stores students as connected nodes

Search

Find any student by ID quickly

CSV File

Saves & loads data from disk

Sort

Display sorted A→Z by name

CRUD

Add, Remove, Search, Update

Loop

Menu runs until user exits (6)

Thank You